

ST PAUL ISLAND, AK, USA

WMO: 703080

Lat: 57.155N

Lon: 170.222W

Elev: 11

StdP: 101.20

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-16.9	-14.6	-20.0	0.6	-16.7	-17.6	0.8	-14.2	18.2	-5.1	16.8	-3.5	7.2	10	0.965

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	13.0	11.4	12.2	10.8	11.4	10.3	11.9	12.5	11.2	11.8	10.6	11.1	5.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	8.4	12.1	10.7	8.0	11.5	10.1	7.7	10.9	33.8	12.6	32.2	11.8	30.8	11.2	15.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	15.2	13.8	DB	-17.7	15.4	4.0	2.3	-20.6	17.0	-22.9	18.4	-25.2	19.7	-28.1	21.4	(4)
(5)			WB	-18.1	12.6	4.0	1.4	-21.0	13.7	-23.3	14.5	-25.6	15.3	-28.4	16.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-3.7	-3.4	-3.8	-1.2	2.4	6.0	8.6	9.6	7.9	4.1	1.2	-1.5	(6)	
	DBStd	5.81	4.83	5.11	5.00	3.47	2.37	1.84	1.65	1.34	1.86	2.43	2.81	3.81	(7)	
	HDD10.0	2863	424	375	428	335	234	121	49	24	67	183	265	358	(8)	
	HDD18.3	5884	683	608	687	585	492	370	302	271	314	441	515	616	(9)	
	CDD10.0	20	0	0	0	0	0	1	6	11	2	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.9	8.2	8.2	7.6	7.2	6.1	5.4	5.0	5.5	6.5	7.3	8.1	8.0	(14)	
(15) Precipitation	PrecAvg	592	44	32	31	31	29	33	49	74	71	77	73	56	(15)	
	PrecMax	897	127	85	83	82	71	78	109	140	144	158	195	118	(16)	
	PrecMin	249	7	0	2	2	3	0	9	4	16	24	22	16	(17)	
	PrecStd	119	23	18	20	18	14	19	23	27	30	30	31	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.1	4.2	5.8	9.7	13.0	14.7	14.2	12.8	10.1	7.3	5.6	(19)	
		MCWB	3.3	3.5	3.1	4.2	6.5	9.7	12.4	12.3	11.3	9.0	6.2	4.8	(20)	
	2%	DB	3.3	3.3	3.3	4.6	7.6	10.8	13.0	13.0	11.7	9.3	6.6	4.5	(21)	
		MCWB	2.7	2.6	2.4	3.4	5.7	8.6	11.3	11.6	10.4	8.2	5.8	3.8	(22)	
	5%	DB	2.7	2.6	2.5	3.8	6.5	9.7	11.9	12.3	11.0	8.4	6.0	3.9	(23)	
		MCWB	2.1	1.9	1.7	2.8	5.1	7.9	10.7	11.1	9.9	7.2	5.2	3.1	(24)	
	10%	DB	2.1	2.1	1.7	2.9	5.6	8.8	11.0	11.6	10.3	7.5	5.2	3.2	(25)	
		MCWB	1.4	1.4	0.9	2.1	4.5	7.3	10.1	10.7	9.3	6.4	4.3	2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.8	3.6	4.7	7.1	10.4	12.8	12.9	11.9	9.6	6.8	5.1	(27)	
		MCDB	3.7	4.0	3.9	5.3	8.8	12.4	14.1	13.8	12.4	9.8	7.0	5.4	(28)	
	2%	WB	2.8	2.8	2.6	3.8	6.1	9.0	11.7	12.0	10.8	8.7	6.1	4.0	(29)	
		MCDB	3.1	3.1	3.0	4.3	7.2	10.4	12.6	12.6	11.2	9.2	6.4	4.3	(30)	
	5%	WB	2.2	2.1	1.8	3.0	5.3	8.2	10.8	11.4	10.3	7.7	5.4	3.3	(31)	
		MCDB	2.6	2.5	2.3	3.6	6.2	9.3	11.5	12.0	10.7	8.3	5.9	3.7	(32)	
	10%	WB	1.5	1.4	1.0	2.2	4.7	7.7	10.2	10.8	9.8	6.8	4.5	2.6	(33)	
		MCDB	1.9	1.9	1.5	2.8	5.5	8.7	10.9	11.3	10.2	7.4	5.1	3.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.9	4.2	3.8	4.0	4.1	3.4	3.3	3.7	3.8	3.5	3.9	(35)	
		MCDBR	2.8	3.2	3.8	3.9	5.3	5.6	4.9	4.1	4.0	3.5	3.5	3.5	(36)	
	5% WB	MCWBR	2.9	3.3	3.6	3.3	3.8	3.8	3.5	3.1	3.2	3.2	3.8	3.6	(37)	
		MCWBR	2.7	3.2	3.6	3.5	4.9	4.3	3.6	3.1	3.0	3.4	3.4	3.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.336	0.332	0.400	0.408	0.403	0.392	0.388	0.370	0.338	0.325	0.271	(40)	
		taud	2.211	2.205	2.287	2.097	2.201	2.306	2.402	2.412	2.359	2.319	2.145	2.320	(41)	
	Edn at Noon		563	693	819	803	827	837	840	817	770	688	498	518	(42)	
		Edh at Noon	52	81	96	136	130	119	107	99	90	71	55	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.40	1.01	2.35	3.53	4.14	4.28	3.69	3.05	2.38	1.39	0.51	0.26	(44)	
	RadStd		0.05	0.11	0.38	0.40	0.30	0.31	0.28	0.31	0.18	0.09	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page